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Project Presentation of Industry Oriented Hands on Experience (22CS422)

On

Illusions of Performance: Fraud Detection using Machine Learning

Geetika, Tanya

2210991581 , 210994849

Supervised By

Swati Goel , Assistant Professor

Department of Computer Science and Engineering,

Chitkara University, Punjab

Introduction

The rapid growth of digital payment systems, online banking, and e-commerce platforms has increased the convenience of financial transactions worldwide. However, this digital transformation has also led to a rise in fraudulent activities such as credit card fraud, identity theft, and unauthorized transactions. rule-based fraud detection systems often fail to detect evolving fraud patternsTraditional effectively. To address this challenge, this project proposes a Machine Learning-based Fraud Detection System that uses algorithms such as Logistic Regression, Random Forest, SVM, and Isolation Forest to identify suspicious transactions. The system also handles class imbalance using cost-sensitive learning and evaluates performance using metrics like Accuracy, Precision, Recall, F1-Score, and ROC-AUC.

Objectives

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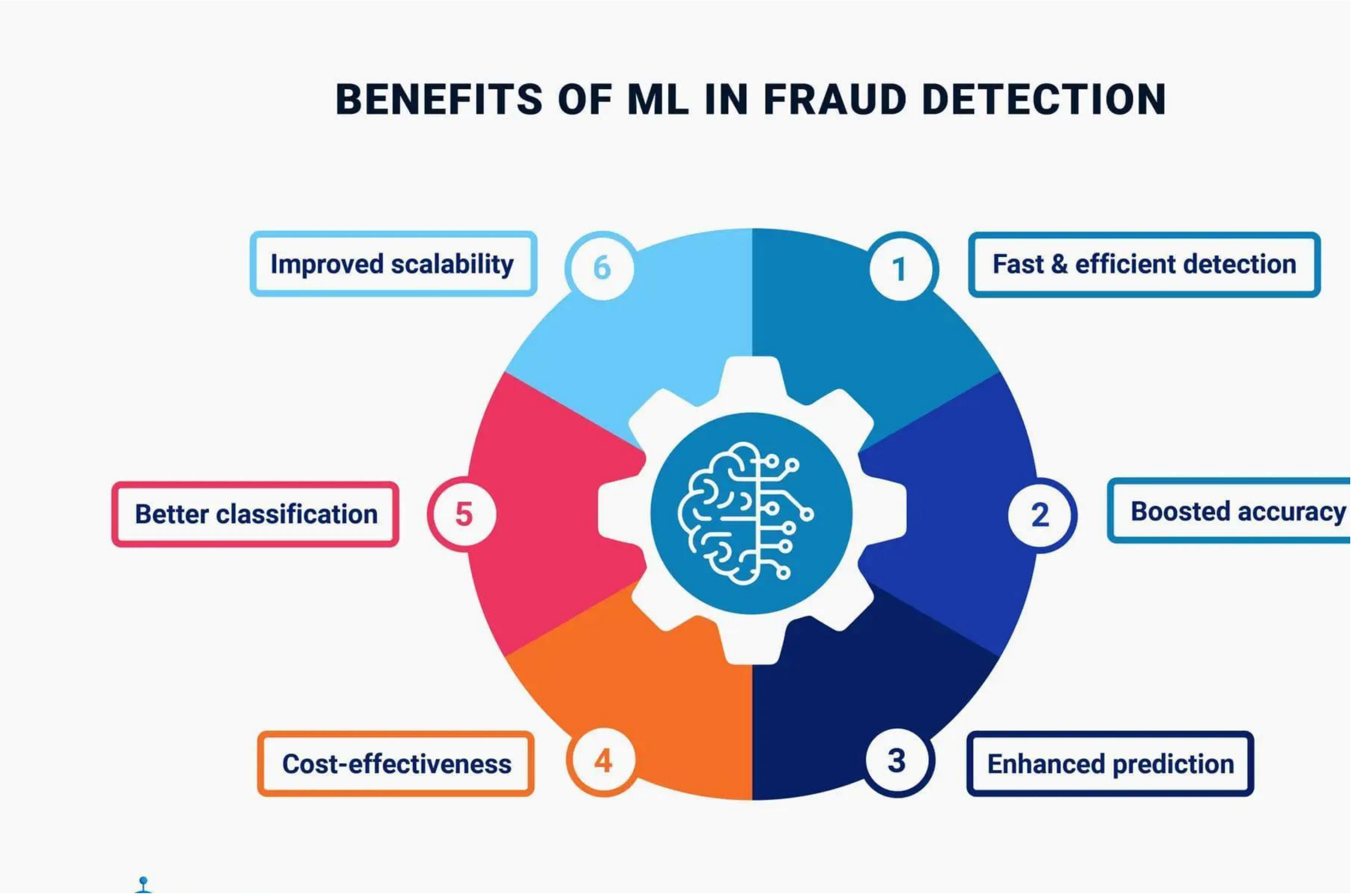
- To develop a machine learning-based fraud detection system for identifying fraudulent financial transactions.
- To implement and compare multiple machine learning algorithms such as Logistic Regression, Decision Tree, Random Forest, SVM, and Isolation Forest.
- To apply cost-sensitive learning techniques for improving fraud detection without modifying dataset distribution.
- To use anomaly detection methods for identifying unusual transaction patterns and previously unseen fraud activities.
- To evaluate model performance using metrics such as Accuracy, Precision, Recall, F1-Score, and ROC-AUC.
- To identify the most effective and robust model for real-time fraud detection in financial systems..

Benefits


BENEFITS OF ML IN FRAUD DETECTION
1
Fast & efficient detection
2
Boosted accuracy
3
Enhanced prediction
4
Cost-effectiveness
5
Better classification
6
Improved scalability

Methodology

Methodology

Step 1 —> Data Collection
Collect financial transaction datasets

Step 2 —> Data Preprocessing
Handle missing values
Remove duplicate/noisy data

Step 3 —> Handling Class Imbalance
Fraud cases are very few compared to normal transactions
Apply Cost-Sensitive Learning

Step 4 —> Feature Engineering
Extract important transaction features:
Transaction frequency

Step 5 — >Model Training
Supervised Models
Logistic Regression

Decision Tree
Random Forest
Support Vector Machine (SVM)

Unsupervised Model

Isolation Forest

Step 6 —> Anomaly Detection

Identify hidden fraud patterns using Isolation Forest

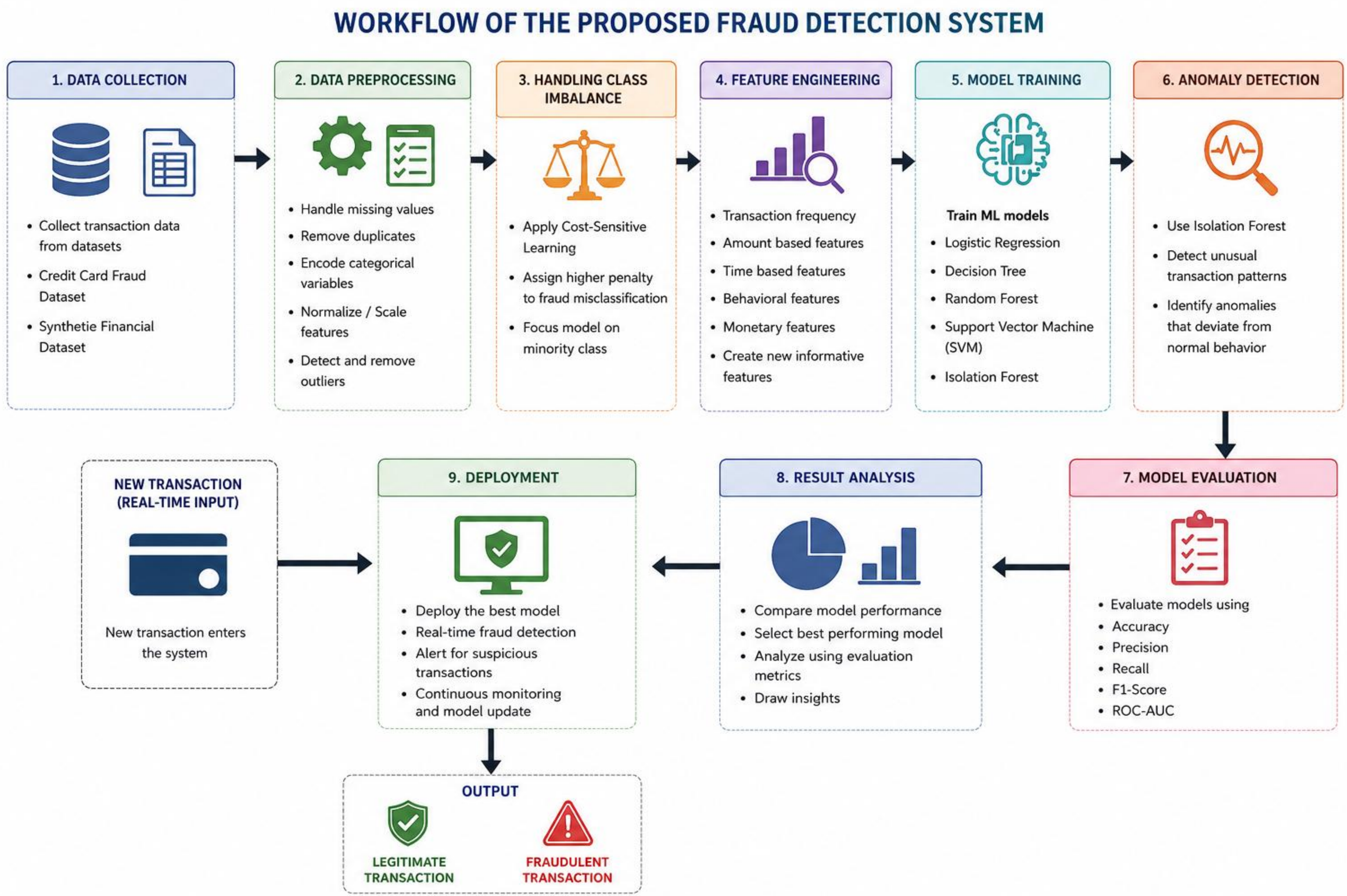
Step 7 —> Model Evaluation

Evaluate performance using:
Accuracy
Precision
F1-Score
ROC-AUC Score

Step 8 —> Result Analysis

Compare model performances
Analyze fraud detection capability
Select best-performing

System WorkFlow



Dataset

The project uses a Credit Card Fraud Detection Dataset containing financial transaction records.

Dataset Features:

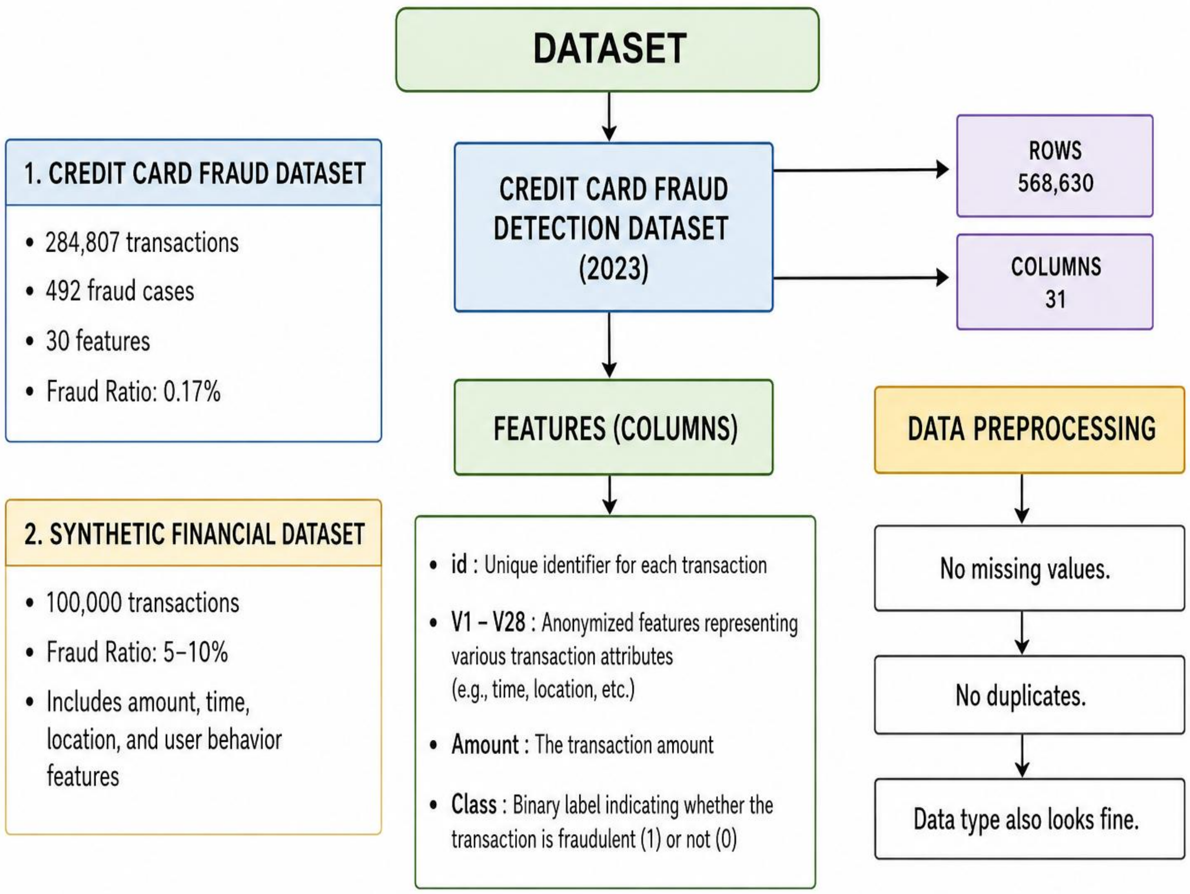
1. Transaction Time
2. Transaction Amount

PCA-based anonymized Features -(V1 to V28)

Class Label

0 → Legitimate Transaction

1 → Fraudulent Transaction



Models Used

- **Logistic Regression**

Used for detecting fraudulent transactions using linear relationships in transaction data.

- **Decision Tree**

Used to classify transactions by splitting data based on fraud-related conditions.

- **Random Forest**

Used to improve fraud detection accuracy by combining multiple decision trees.

- **Support Vector Machine (SVM)**

Used for identifying complex fraud patterns in high-dimensional transaction data.

- **Isolation Forest**

Used for anomaly detection by identifying unusual transaction behaviors.

System Requirements

Hardware Requirements:

Component	Specification
Processor	Intel Core i5 / Ryzen 5 or above
RAM	Minimum 8 GB
Storage	256 GB SSD or higher
GPU	Optional for advanced model training
Internet Connectivity	Required for dataset access and research

Software Requirements:

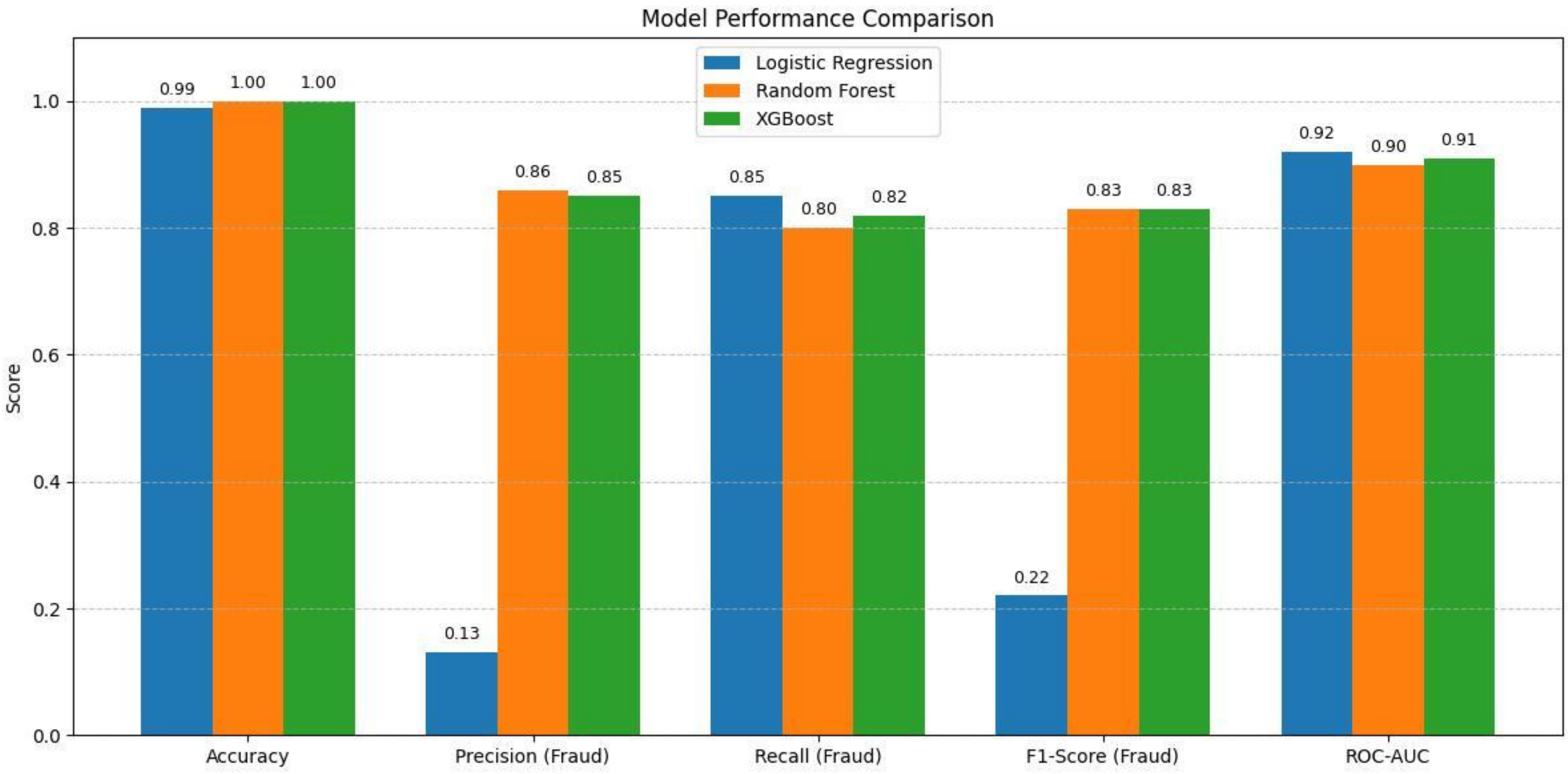
Software	Purpose
Python 3.x	Programming Language
Jupyter Notebook / VS Code	Development Environment
Scikit-learn	Machine Learning Models
Pandas	Data Processing
NumPy	Numerical Computation
Matplotlib & Seaborn	Data Visualization
Operating System	Windows / Linux / macOS

Evaluation Metrices

- **Accuracy**
  Measures overall correct predictions made by the model.
- **Precision**
  Measures how many predicted fraud transactions are actually fraud.
- **Recall**
  Measures how many fraud transactions are correctly detected.
- **F1-Score**
  Provides a balance between Precision and Recall.
- **ROC-AUC**
  Measures model's ability to distinguish between fraud and legitimate transactions.

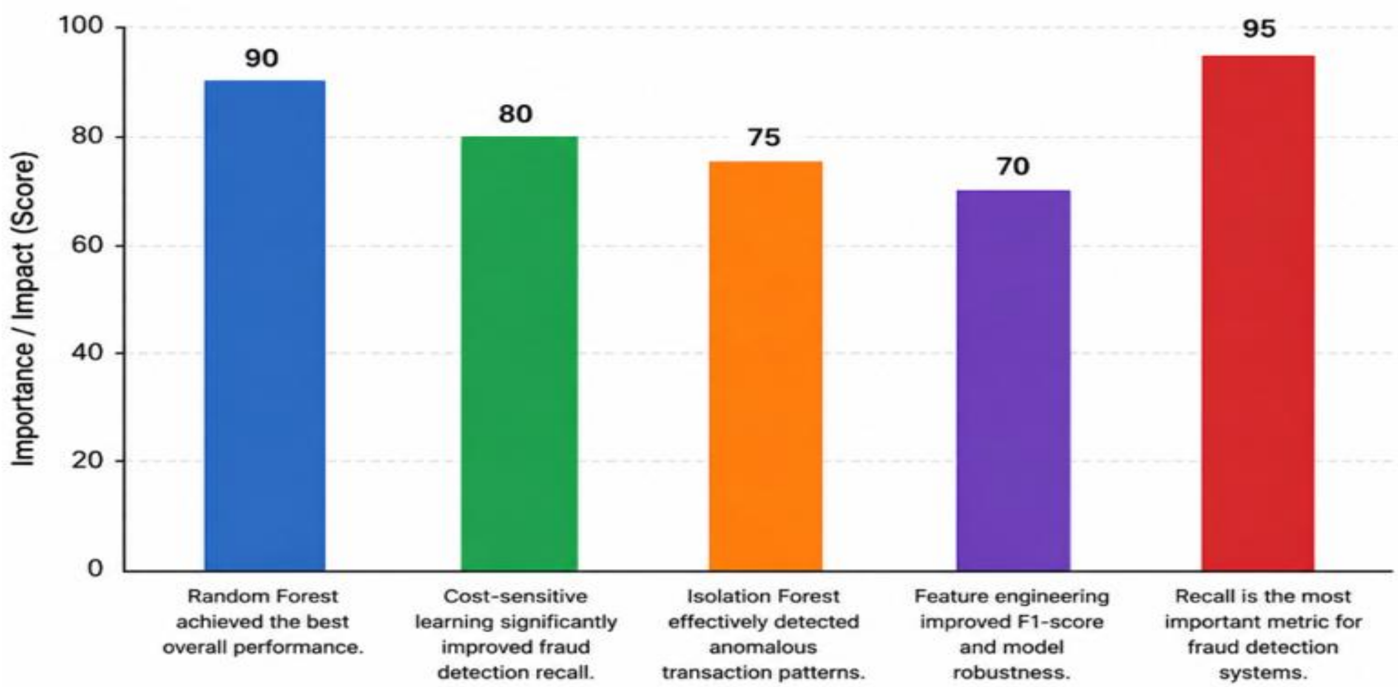
Recall and F1-score are prioritized due to the highly imbalanced nature of fraud datasets.

Model Performance Comparison



Key Findings

- Random Forest achieved the best overall performance.
- Cost-sensitive learning significantly improved fraud detection recall.
- Isolation Forest effectively detected anomalous transaction patterns.
- Feature engineering improved F1-score and model robustness.
- Recall is the most important metric for fraud detection systems.



Skill Gap Analysis

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- Identifies missing skills in a candidate profile
- Compares user skills with industry requirements
- Helps improve career recommendations
- Suggests skills for improvement and learning
- Enhances employability and job readiness

Conclusion

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This project demonstrates that Machine Learning techniques combined with feature engineering, anomaly detection, and cost-sensitive learning can significantly improve fraud detection performance in highly imbalanced financial datasets. The implemented framework provides a scalable and practical solution for modern fraud detection systems .The proposed system can support real-time fraud monitoring systems

Future Scope

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- Add real-time job data
- Use deep learning models
- Improve recommendation accuracy
- Improve detection accuracy using advanced AI models
- Integrate Explainable AI for better transparency
- LSTM and Transformer-based models
- Explainable AI for fraud prediction

Thank You